

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 7. String and its Methods

Lecture: 82. URL Parsing- Project

Section 1: Lecture Summary

The lecture focused on the concept of URL parsing, using a project-based approach in Python. It began by explaining what a URL (Uniform Resource Locator) is, describing its components: protocol (e.g., http, https), domain name (e.g., kaggle), and page or query string (e.g., /datasets). The session showed how to extract these elements from a URL string by performing string operations such as finding character positions and slicing substrings. The instructor demonstrated breaking down the URL "https://www.kaggle.com/datasets" into its components and guiding through writing a Python program to accept a URL input and output the protocol, domain name, and page. The focus was on string methods like `find()` and slicing for parsing each part effectively.

Section 2: Key Concepts and Explanations

- URL and Its Components:

- URL (Uniform Resource Locator) or URI (Uniform Resource Indicator) identifies the address of a resource on the Internet.

- Typical URL structure: protocol://www.domain.extension/page

- Protocol: Specifies the communication protocol (e.g., http, https, ftp).

- Domain Name: The name identifying the website, found between two dots after www.

- Page or Query String: The path or location after the domain name, starting from the slash (/).

- Parsing a URL:

- Parsing means extracting meaningful components from the URL string.

- Components to extract: protocol, domain name, and page.
- Methods used:
 - `find(character)` to locate positions of special characters like ':', '.', '/'.
 - String slicing `[start:end]` to extract substrings based on indices.
- Strategy:
 1. Find the colon (:) to extract the protocol from the start up to the colon.
 2. Find the first and second dot (.) indices to extract the domain name (between the two dots).
 3. Find the slash (/) after the domain part to extract the page path including the slash.
- Python String Operations Relevant for Parsing:
 - `find(substr)` returns the lowest index of substring if found, else -1.
 - String slicing syntax: `string[start:end]`, extracts substring starting at 'start' index up to but not including 'end' index.
 - Slicing with omitted start or end defaults to beginning or end of the string.
 - Using offsets in `find(substr, start)` to search from a specific position.

Section 3: Example Code and Use Cases

Sample program to parse URL into protocol, domain, and page

Extract protocol

Find indices of dots for domain extraction

Extract domain (between first and second dot)

Extract page - substring from slash '/' occurring after second dot

```
url = "https://www.kaggle.com/datasets"

protocol = url[:url.find(":")]

dot1 = url.find(".")
```

```
dot2 = url.find(".", dot1 + 1)

domain = url[dot1 + 1:dot2]

page = url[url.find("/", dot2):]

print("Protocol:", protocol)
print("Domain:", domain)
print("Page:", page)
```

Explanation:

- The program takes a URL string hardcoded here but can be adapted for user input.
- It finds the protocol by slicing from the start to the colon's index.
- It finds the first and second periods to locate the domain portion between them.
- It finds the slash after the domain to extract the page path starting from that slash.
- Finally, it prints all extracted parts.

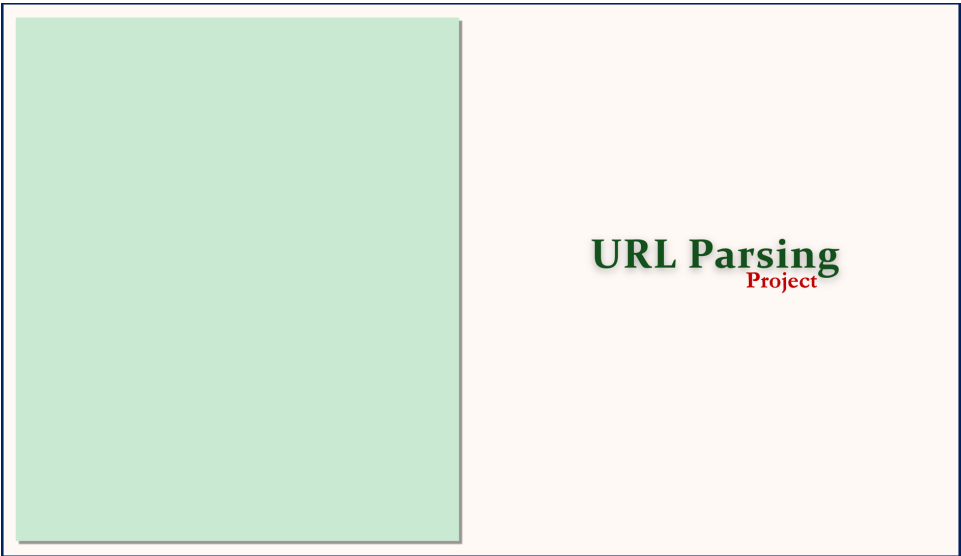
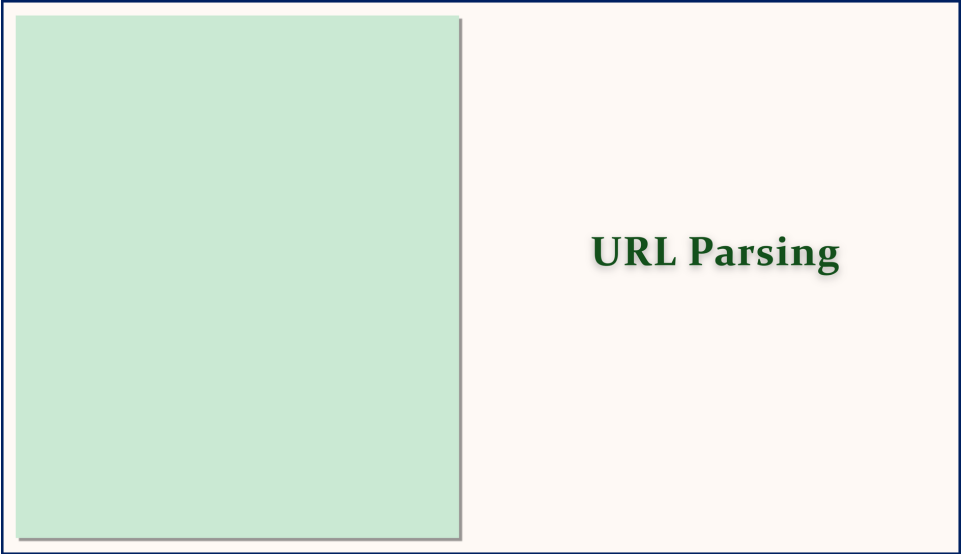
Section 4: Key Takeaways

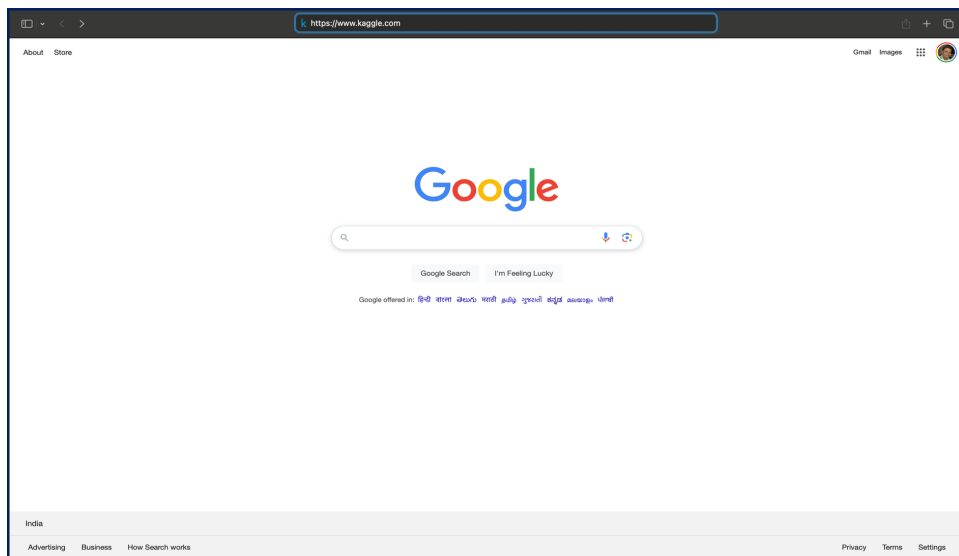
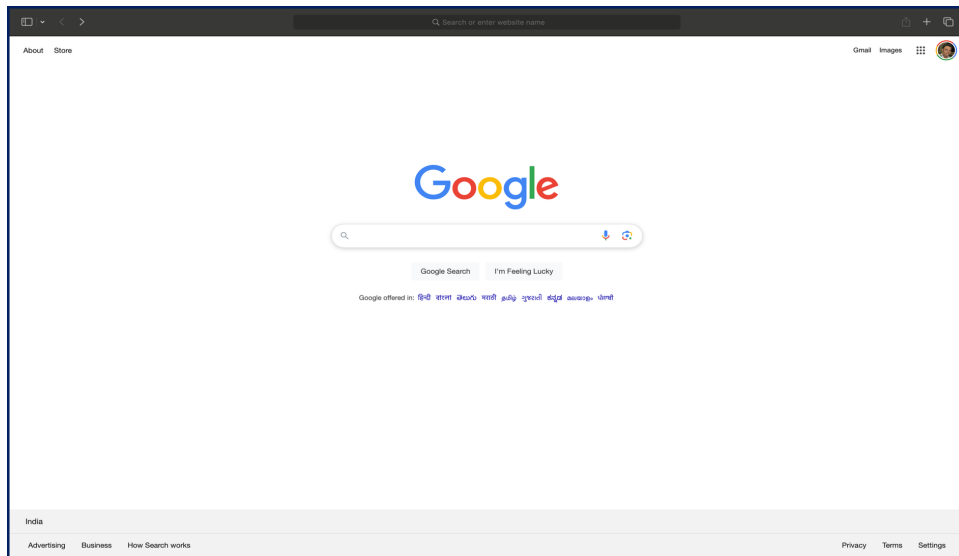
- A URL consists mainly of protocol, domain name, and page/path components.
- Parsing a URL means extracting these components from the URL string.
- String methods like `find()` and slicing are powerful tools for parsing strings in Python.
- Knowing the structure of the URL helps in strategically locating characters to slice substrings.
- This project reinforces practical use of string operations to build a useful URL parsing program.

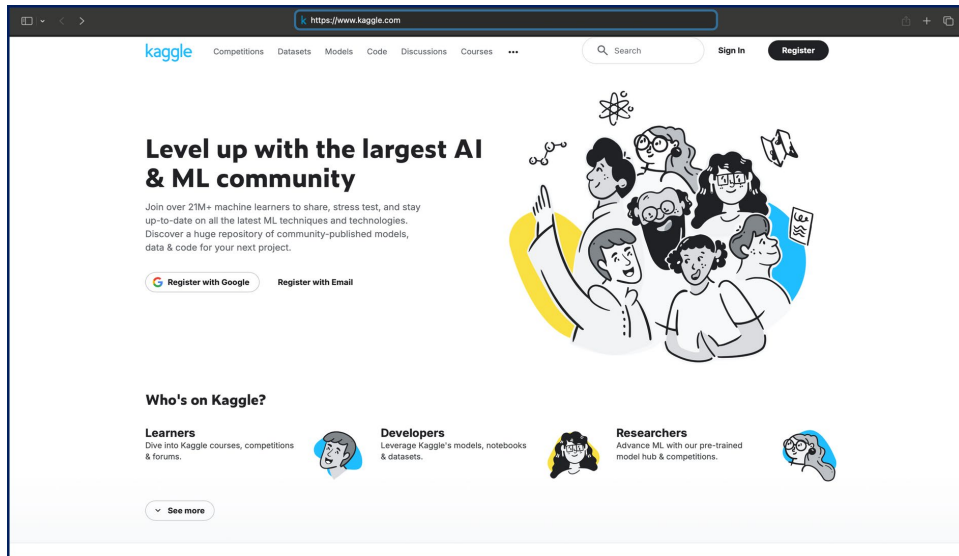
Lecture in Slides

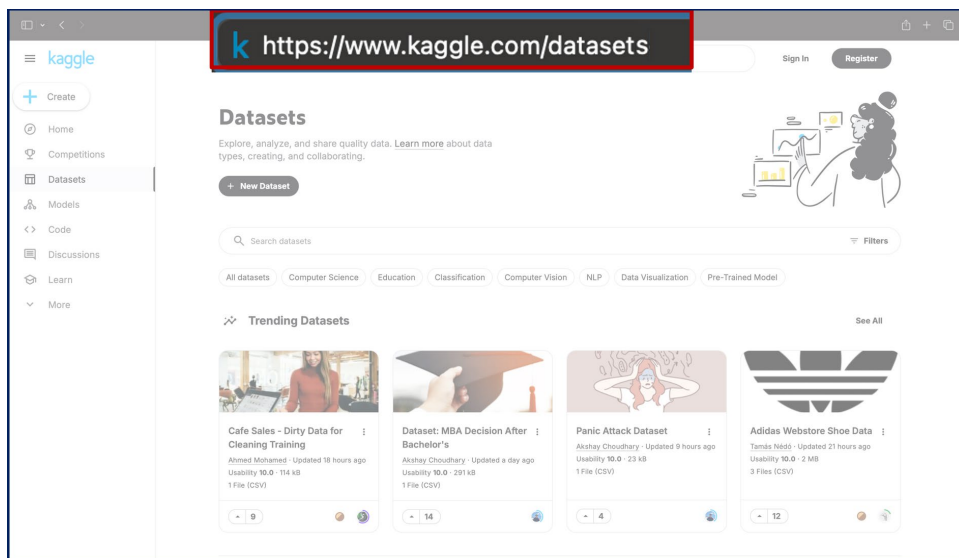
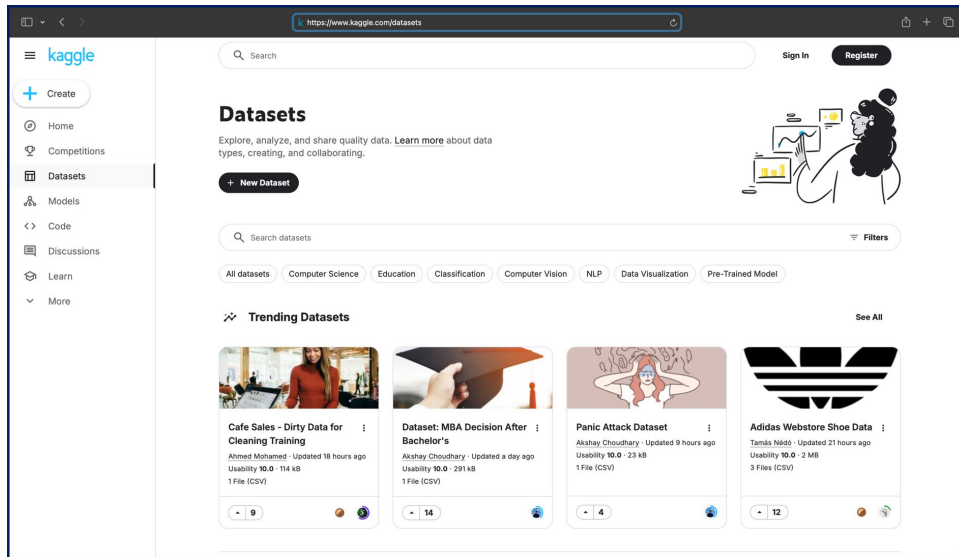
Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.

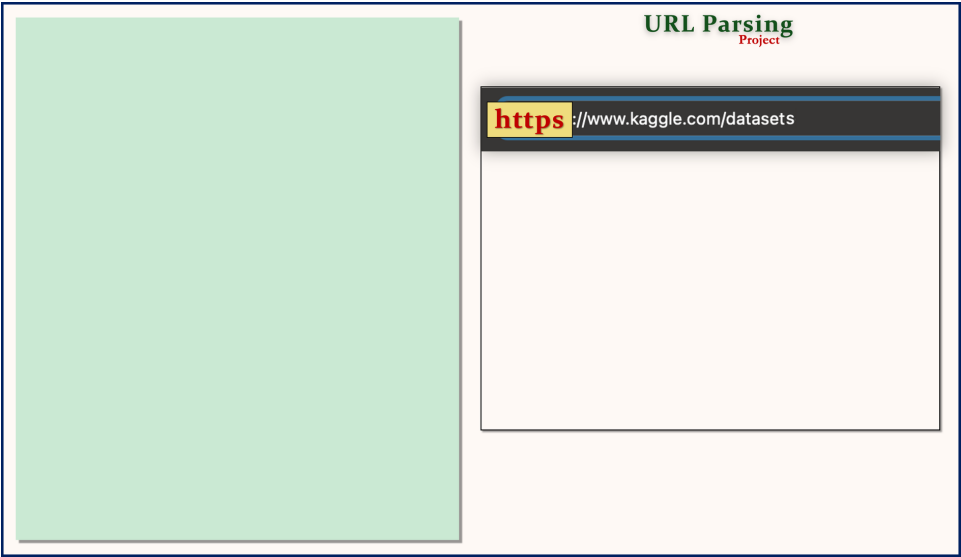
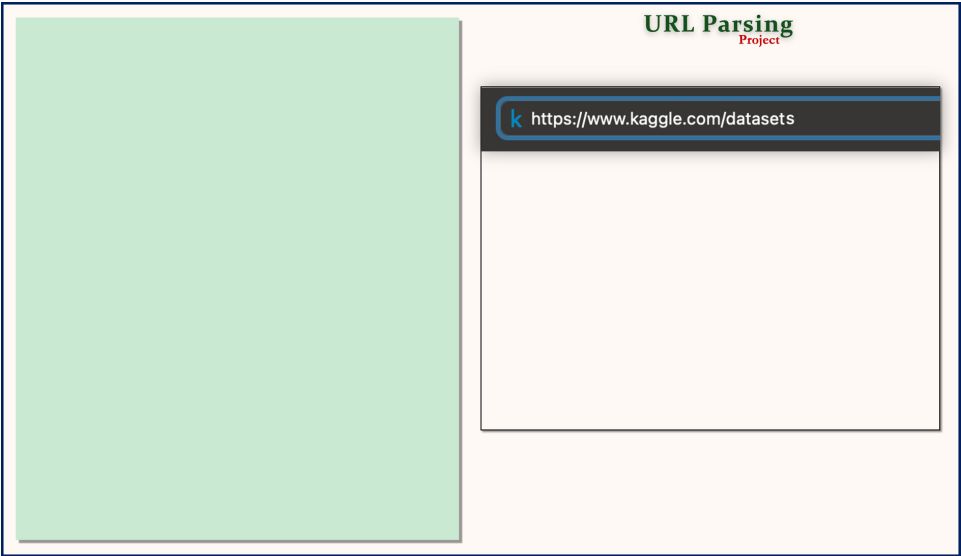


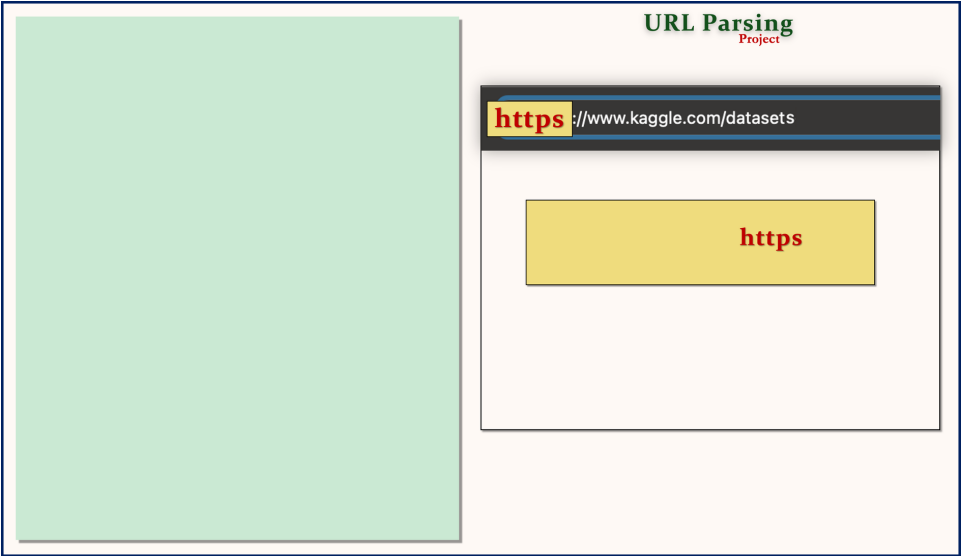












URL Parsing
Project

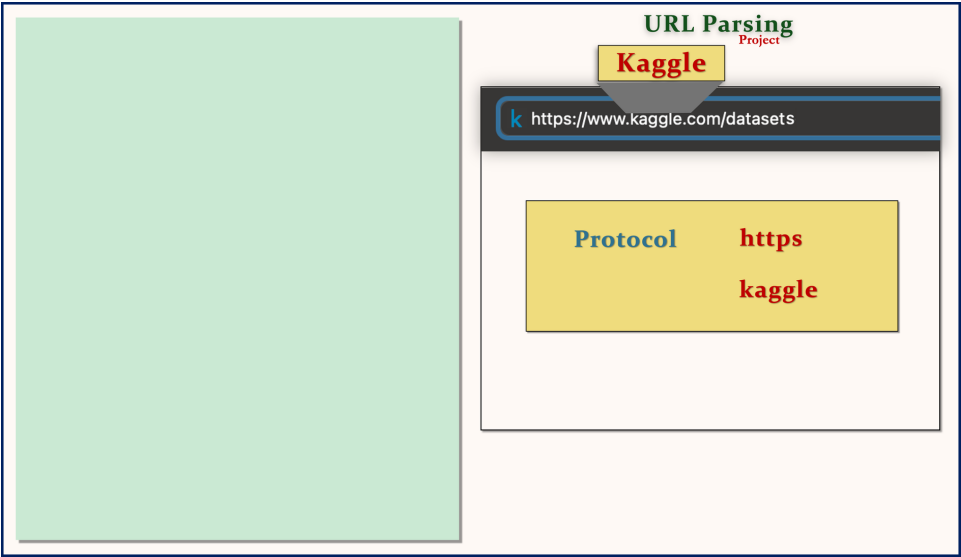
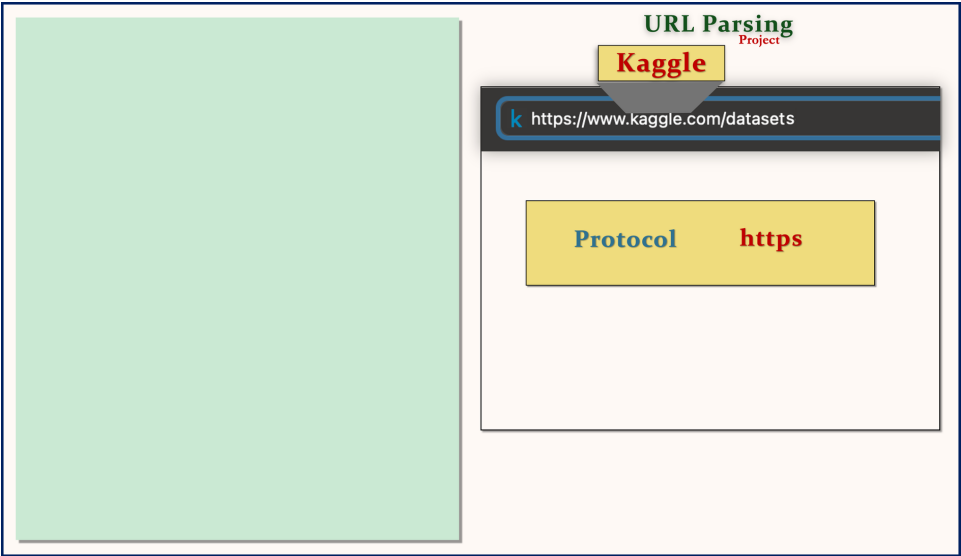
https://www.kaggle.com/datasets

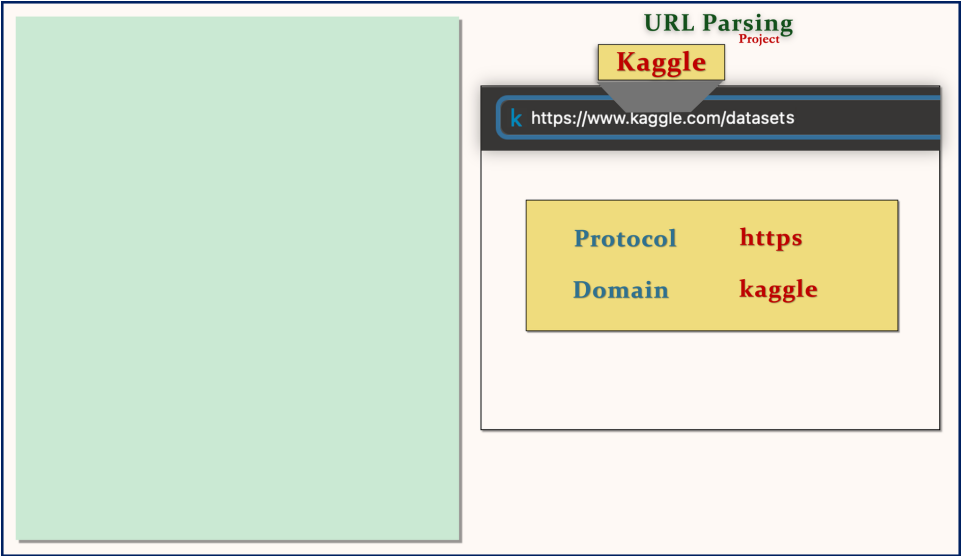
Protocolhttps

URL Parsing
Project

k https://www.kaggle.com/datasets

Protocolhttps





URL Parsing
Project

k https://www.kaggle.com/datasets

Protocol	https
Domain	kaggle

URL Parsing
Project

k https://www.kaggle.com/ datasets

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URL Parsing
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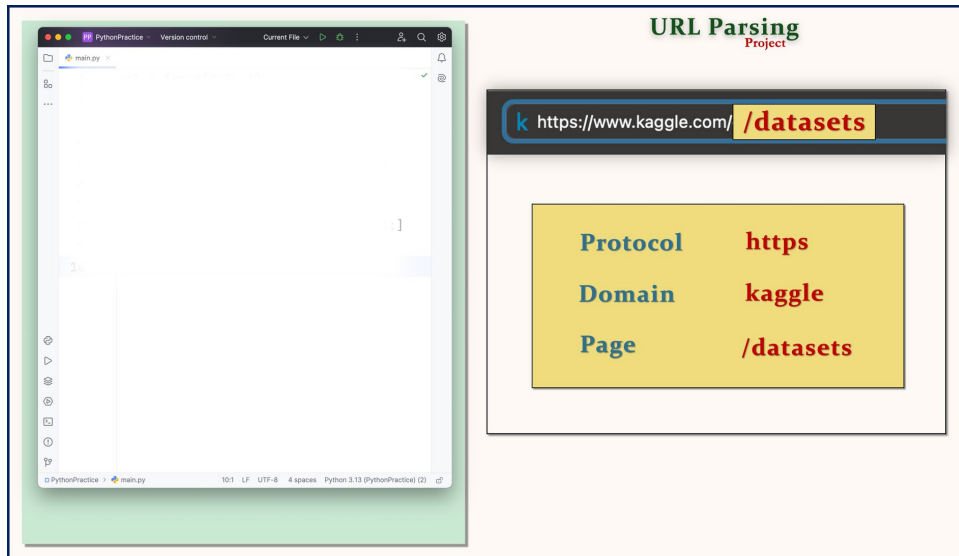
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Protocol	https
Domain	kaggle
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URL Parsing
Project

k https://www.kaggle.com/ /datasets

Protocol	https
Domain	kaggle
Page	/datasets



URL Parsing Project

k <https://www.kaggle.com/> **/datasets**

Protocol	https
Domain	kaggle
Page	/datasets

The screenshot shows the PyCharm IDE with a file named `main.py`. The code in the editor is:

```
url = "https://www.kaggle.com/"
```

The Run console shows the output:

```
main.py:1: URL |
```

URL Parsing Project

k <https://www.kaggle.com/> **/datasets**

Protocol	https
Domain	kaggle
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The Run console shows the output:

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main.py:1: URL |
main.py:1: Protocol http
main.py:1: Domain kaggle
main.py:1: Page /datasets
main.py:1: Process finished with exit code 0
```

URL Parsing

Project

k https://www.kaggle.com/datasets

protocoldomainpage

URL Parsing

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k https://www.kaggle.com/datasets

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URL Parsing

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URL Parsing

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URL Parsing

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URL Parsing

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Protocol https

Domain kaggle

Page /datasets